Network Science

Lecture 06: Structural Balance

5942UE Network Science

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Structural Balance

We now study **signed networks** where relations can be positive or negative.

Focus

- complete signed graphs
- structural balance and its global theorem
- applications to politics and online trust
- weak and relaxed balance

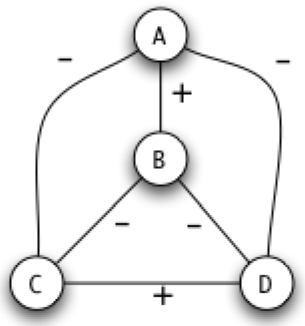
Module 1

Structural Balance Theorem

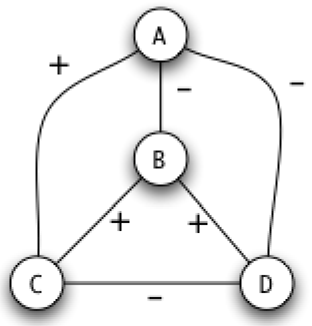
Guiding Question

What global network structures are compatible with psychologically plausible signed triangles?

Structural Balance Property



balanced



not balanced

A complete signed graph is **structurally balanced** if every triangle is either all-positive or has exactly one positive edge.

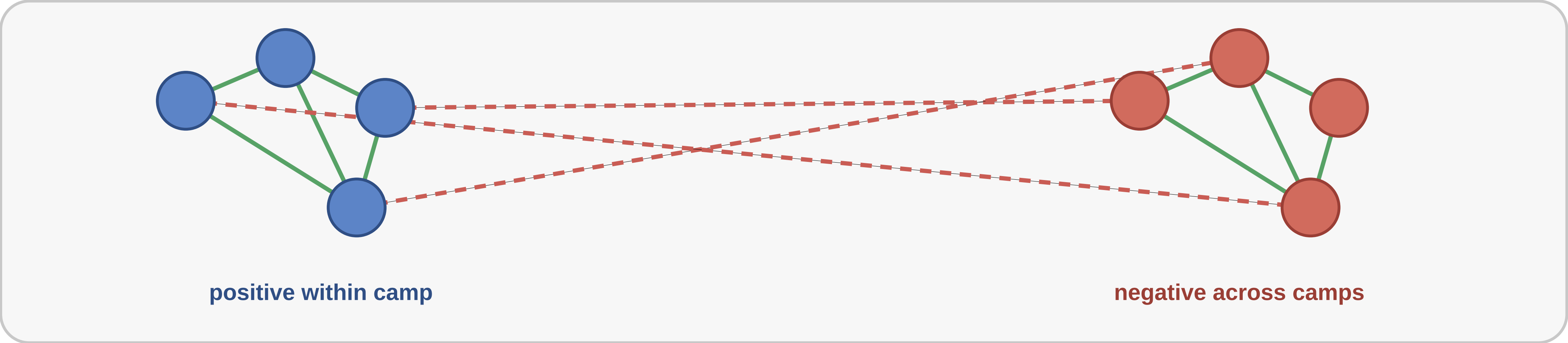
Core Result

If a complete signed graph is balanced, then its nodes can be partitioned into:

- one all-positive block
- or two internally positive blocks with all negative edges between them

Balanced as Camps

Balanced signed structure collapses into camps



Interpretation

If the graph is balanced, it must collapse into either:

- one all-friends block
- or two internally friendly blocks with hostility between them

> This is a classic example of a local constraint producing strong global organization.

Module 2

Applications

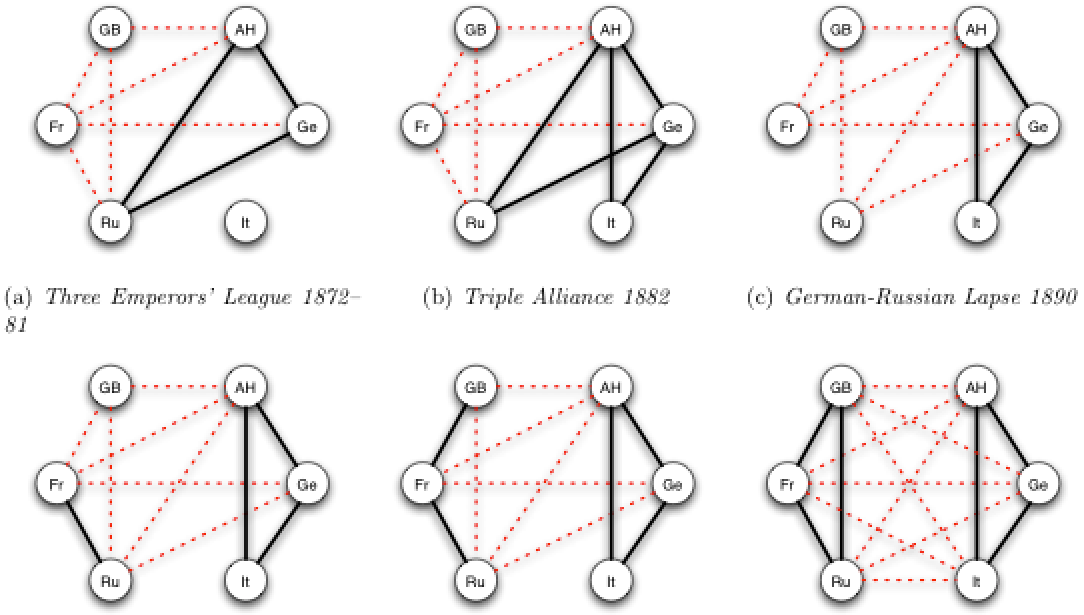
Guiding Question

What do signed-network models help us explain in real settings?

Formal Intuition

Signed networks extend graphs by giving each edge a **positive or negative label**.

International Relations



(a) *Three Emperors' League 1872–81*
(b) *Triple Alliance 1882*
(c) *German-Russian Lapse 1890*
(d) *French-Russian Alliance 1891–94*
(e) *Entente Cordiale 1904*
(f) *British Russian Alliance 1907*

Structural balance can describe alliance formation and polarization into camps.

Online Trust

Signed ties can also model:

- **trust and distrust**
- **recommendation and warning**
- **support and opposition**

Interpretation

On trust platforms, positive and negative links do not behave exactly like undirected friendship and hostility, but the balance framework still suggests useful questions:

- when should distrust propagate?
- when do trust triangles stabilize?
- what patterns indicate factional structure?

> Applications are strongest when the signed model is treated as an approximation, not as a literal truth

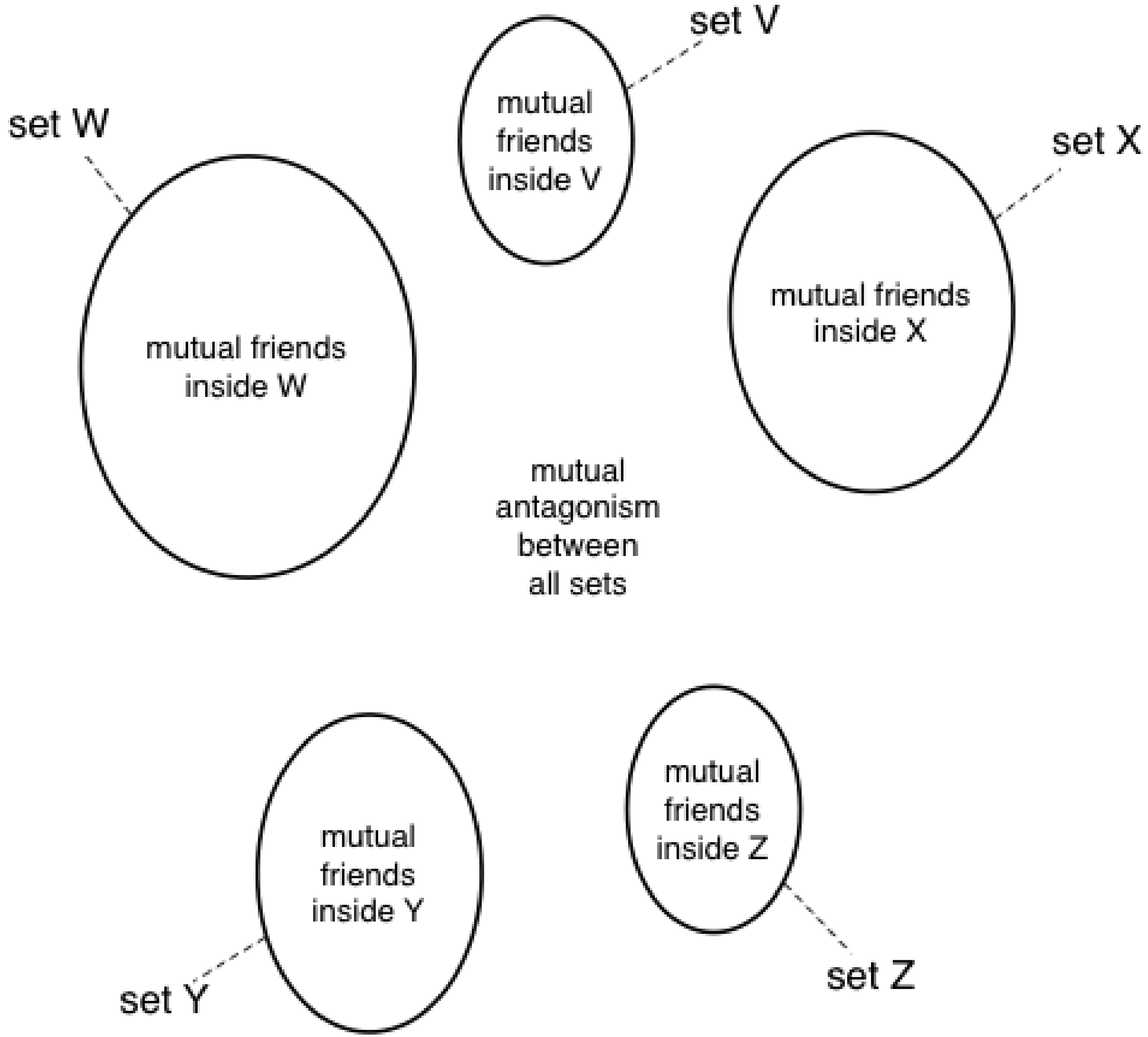
Module 3

Weak Structural Balance

Guiding Question

What changes if "the enemy of my enemy" is not forced to become my friend?

Counterexample


set V
mutual friends inside V
set W
mutual friends inside W
set X
mutual friends inside X
mutual antagonism between all sets
mutual friends inside Y
set Y
mutual friends inside Z
set Z

Weak Balance

Weak structural balance forbids triangles with exactly two positive edges and one negative edge, but allows all-negative triangles.

Speaker notes

State the rule clearly before the interpretation. The point is that one local relaxation changes the global theorem.

Interpretation

That relaxation changes the global theorem:

- instead of exactly two camps
- the graph may split into **multiple** internally positive groups
- with negative ties between groups

> Changing one local rule can change the entire global picture.

Module 4

Relaxed and Incomplete Balance

Guiding Question

How can we reason about balance when graphs are incomplete or only approximately balanced?

Incomplete Graphs

Not every pair of actors has a known signed relation. That forces us to ask whether balance can still be inferred from **partial evidence**.

Odd Negative Cycles

$$\text{odd number of negative edges on a cycle} \Rightarrow \text{not balanced}$$

Formal Intuition

Once completeness is dropped, balance is no longer a simple triangle-only condition.

Instead, the question becomes whether the observed graph is compatible with an **approximately balanced partition**.

Approximate Balance

In real data, a graph can be **mostly balanced** without satisfying the theorem perfectly. Then the right question is whether it is close to a balanced partition.

> Relaxed balance is more realistic, but it replaces sharp theorems with softer structural diagnostics.

Wrap-Up

- local triangle constraints imply strong global structure
- signed networks are analytically powerful but assumption-heavy
- relaxing the assumptions changes both the theorem and the interpretation